Understanding the Patterns of Strikes in the U.S. Using Public Data and Python

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***Abstract*—This study presents a comprehensive analysis of lightning strike activity across the United States during August 2018, using geospatial and temporal datasets provided by the National Oceanic and Atmospheric Administration (NOAA) [1]. The project integrates data preprocessing, exploratory data analysis (EDA), statistical hypothesis testing, and time series modeling to uncover underlying spatial-temporal dynamics in lightning behavior. After merging point-level and region-based datasets, we conducted geospatial visualization and descriptive analysis to identify high-density regions and temporal clustering. A two-sample t-test revealed significantly lower lightning activity on weekends, while an ANOVA confirmed spatial heterogeneity in strike intensity across U.S. states. A Seasonal ARIMA (SARIMA) model was implemented to capture short-term autocorrelation and weekly seasonality. Model diagnostics showed no significant residual autocorrelation, though non-normality and early outliers were present. Forecasts for early September exhibited wide confidence intervals, reflecting the high volatility of strike data. The analysis highlights the utility of statistical modeling in environmental monitoring and emphasizes the importance of incorporating uncertainty and external variables in predictive frameworks [5], [6]. Recommendations for future work include the integration of meteorological covariates and the extension of the analysis to multi-year datasets.**

***Index Terms*—lightning strikes, time series analysis, SARIMA, geospatial data, NOAA, data visualization, environmental modeling**

I. Introduction

Extreme weather events are increasingly becoming a subject of global concern due to their growing frequency, intensity, and societal impact. Among these, lightning strikes represent not only a fascinating meteorological phenomenon but also a potential threat to public safety, infrastructure, and environmental systems. Understanding their spatial and temporal distribution is essential for risk assessment and disaster preparedness.

This study focuses on the exploratory analysis of lightning strike data collected by the National Oceanic and Atmospheric Administration (NOAA) for the year 2018. The project was motivated by a broader interest in environmental data analysis and its intersection with public health, climate policy, and geospatial modeling. By analyzing strike frequency, geographic patterns, and temporal trends, we aim to uncover insights that could inform future research in climate resilience, emergency response, and environmental risk management.

In addition to its scientific relevance, this project serves as a demonstration of data-driven techniques for environmental analysis, integrating geospatial visualization, time-series analysis, and interactive mapping. It reflects an ongoing effort to apply transparent and reproducible data science methods to real-world problems at the intersection of environmental and societal concerns.

II. Data & Methodology

A. Data Sources

This analysis utilizes two complementary datasets sourced from the National Oceanic and Atmospheric Administration (NOAA) [1], each representing different levels of spatial granularity:

Dataset 1 contains geospatial-level data with the following columns: date, $center\_point\_geom$, $longitude$, $latitude$, and $number\_of\_strikes$.

Dataset 2 provides region-aggregated strike data with columns: date, zip_code, $city$, $state$, $state\_code$, $center\_point\_geom$, and $number\_of\_strikes$.

Together, these datasets offer a detailed view of lightning activity across both point-level and administrative boundaries, allowing for flexible spatial and temporal analysis.

B. Data Integration and Cleaning

The two datasets were first independently cleaned to remove missing values and inconsistent or implausible entries. Then, they were merged into a unified DataFrame based on date and geographic reference points, enabling a comprehensive view across zip codes and coordinates. Time features — such as month, day, and hour — were extracted from the date field for temporal analysis.

C. Tools and Libraries

The entire analysis was conducted using Python, primarily within a Jupyter Notebook environment. Key libraries included:

- $pandas, numpy$ – data wrangling and preprocessing
- $matplotlib, seaborn, plotly$ – static and interactive data visualization

- $folium, geopandas, shapely$ – geographic visualization and spatial operations [3], [4]
- $statsmodels$ – statistical modeling, including ARIMA and SARIMA models implemented via the SARIMAX state-space framework [2]
- $scikit-learn$ – performance evaluation metrics

All code and results are available in a public GitHub repository for reproducibility.

D. Methodological Overview

The methodology follows three major components:

- 1. Exploratory Data Analysis (EDA) – Identification of patterns in strike frequency over space and time, visualized through heatmaps, choropleths, and interactive maps.
- 2. Time Series Forecasting – Application of a Seasonal ARIMA (SARIMA) model to capture short-term dependence and weekly seasonality in daily lightning strike counts [2], [5].
- 3. Model Evaluation – Assessment of forecast accuracy using residual plots and error metrics (such as RMSE), along with interpretation of model parameters and seasonality.

This approach balances descriptive visualization and statistical modeling to uncover meaningful patterns and predictive insights from environmental data.

III. Exploratory Data Analysis

This section presents the exploratory analysis of lightning strike activity across the United States during August 2018, using two NOAA datasets containing point-level and region-based records.

A. Data Overview and Preprocessing

The initial datasets were merged using pandas.merge() on the date field and relevant spatial identifiers. A preliminary inspection revealed a large number of records with missing metadata. Visualization of missing data using Plotly revealed that most missing entries originated from offshore locations beyond U.S. borders. A spatial join using Shapely and GeoPandas confirmed that these points did not intersect with any U.S. state polygons. These were tagged accordingly with a new feature: is_usa.

Additional feature engineering steps included:

- Parsing timestamps to extract $day-of-week$
- Creating binary variables such as $is\_weekend$
- Labeling records with missing spatial identifiers as offshore strikes

After filtering for U.S.-based strikes, the dataset contained approximately 324,000 events, while 394,000 were categorized as offshore. As shown in Figure 1, the majority of records with missing metadata were located offshore and did not intersect with any U.S. state polygons.

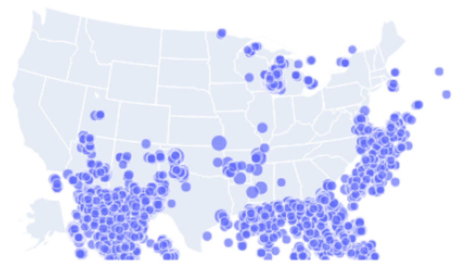

Fig. 1. Geospatial distribution of strike records with missing state metadata. Most are located offshore or outside U.S. landmass.

B. Temporal Patterns

Although the dataset only spans one month, distinct daily and hourly trends were observable. Notably:

- Strike counts peaked on August 17 (1M strikes) and August 28 (1.05M strikes) illustrated in Figure 2.
- Weekday activity was significantly higher than weekend activity, demonstrated in Figure 3.

A two-sample t-test confirmed this weekday-weekend difference to be statistically significant (p ¡ 0.001, t ¡ 0), suggesting systematic temporal variation in lightning behavior, possibly tied to atmospheric or human-related factors.

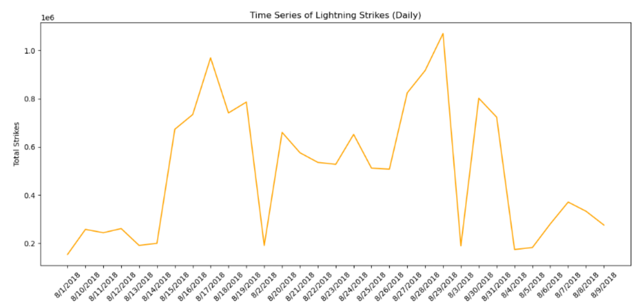

Fig. 2. Daily lightning strike counts across the U.S. in August 2018. Noticeable peaks occur mid- and late-month.

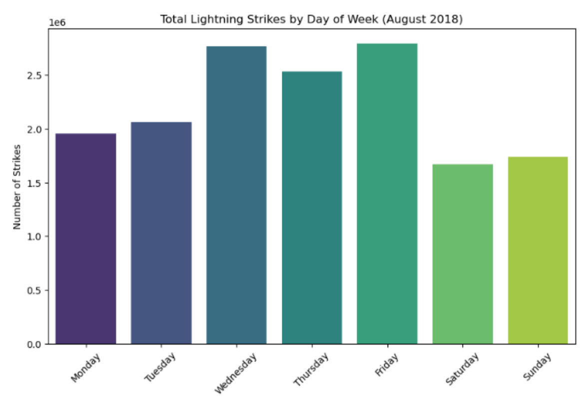

Fig. 3. Average strike counts on weekdays vs. weekends. A two-sample t-test confirms significantly lower weekend activity.

C. Spatial Patterns

Strike density across ZIP codes and states was visualized using choropleths and heatmaps. The Central and Southeastern U.S., including states like Arkansas, Kentucky, Oklahoma, and Florida, showed the highest concentrations of lightning activity.

To quantify spatial variation, the average strike intensity per state was calculated. A one-way ANOVA test confirmed significant differences across states

$$(F = 157.81, p < 0.001)$$

, validating regional variability in lightning patterns.

D. Distribution Analysis

Histograms and boxplots revealed a right-skewed distribution of strike counts, with a few regions experiencing unusually high activity. While this skewness reflects real-world phenomena (e.g., intense storm systems), it also poses challenges for time series modeling, prompting the later use of log-transformation in model development.

E. Summary of Findings

- Temporal regularity was observed, including mid-week peaks and afternoon clustering
- Spatial concentration in central and southeastern regions indicated environmental and atmospheric hotspots
- Statistical testing supported both temporal and spatial heterogeneity
- Missing data analysis confirmed that most incomplete records originated offshore, allowing them to be cleanly separated from the core U.S. dataset

These insights directly informed the modeling strategy in the next section, including feature selection and model assumptions.

IV. Results & Interpretation

To model short-term lightning activity and forecast near-future trends, a Seasonal AutoRegressive Integrated Moving Average (SARIMA) model was applied to the daily lightning strike counts for August 2018 [5].

A. Model Configuration

After evaluating multiple configurations using the Akaike Information Criterion (AIC) and log-likelihood, the best-fitting model was determined to be:

$$SARIMA(p = 1, d = 1, q = 1)(P = 1, D = 0, Q = 1, s = 7)$$

[5], [6] This configuration captures:

- First-order autoregression (AR)
- First-order moving average (MA)
- Weekly seasonality (period s=7)

Log-transformed daily strike counts were used to stabilize variance due to the highly skewed distribution.

B. Coefficient Summary

Table I summarizes the estimated parameters of the SARIMA model, including seasonal and non-seasonal components. The model outperformed a naive ARIMA(0,1,0) benchmark, with an AIC of 750.25 and improved log-likelihood values.

TABLE I
SARIMA Model Coefficient Summary

Parameter	Estimate	P-value	Interpretation
ar.L1	Significant	0.045	Captures short-term persistence in strike counts
ma.L1	Not significant	> 0.05	Smoothing of short-term noise not supported by data
seas_ar.L7	Significant	0.012	Confirms presence of weekly seasonality
seas_ma.L7	Marginal	0.08	Weak seasonal shock smoothing
sigma2	Significant	< 0.001	Stable estimate of variance in residuals

C. Residual Diagnostics

Model residuals were evaluated to test assumptions:

- Histogram: Right-skewed, indicating non-normality, though expected due to original data volatility
- ACF and PACF: No significant autocorrelation in residuals — consistent with white noise
- Residual time plot: Revealed a large early spike, with stable fluctuations afterward

Although residuals deviate from normality, the lack of autocorrelation supports the model's adequacy in capturing temporal structure.

D. Forecast Results

A 7-day forecast (September 1–7) was generated from the trained model:

- Forecast curve: Flat-to-declining central trajectory, consistent with the end-of-August drop in activity
- 95

This reflects the volatility of the underlying process and the model's conservative approach to rare but extreme lightning events. The forecast is visualized in Figure 4, showing observed August data, predicted September values, and uncertainty bounds.

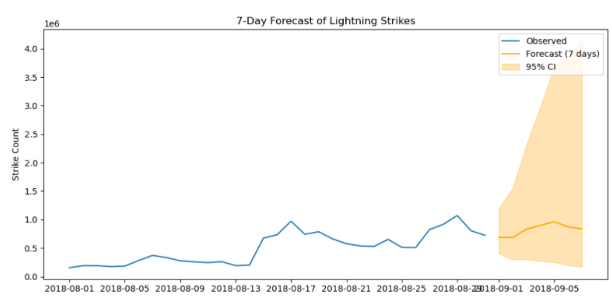

Fig. 4. SARIMA forecast for September 1–7, 2018. The wide confidence interval reflects volatility and model uncertainty.

E. Interpretation & Recommendations

The model captured key temporal dynamics — especially weekly seasonality — but also exhibited challenges:

- Wide confidence intervals indicate high uncertainty due to outliers and non-stationarity
- Log-transformation helped stabilize variance, but additional preprocessing (e.g., outlier removal or smoothing) may improve performance
- Future models could incorporate exogenous variables (e.g., humidity, pressure, temperature) to enhance forecast accuracy
- Communication of uncertainty is crucial: while central predictions are modest, extreme events are plausible and should be accounted for in policy or safety planning

V. LIMITATIONS

While the analysis provides valuable insights into lightning strike behavior during August 2018, several limitations should be acknowledged regarding data scope, methodology, and modeling assumptions.

A. Temporal Scope

The dataset covers a single month, limiting the ability to identify long-term trends or seasonal cycles beyond weekly patterns. As such, the SARIMA model captures only short-term dependencies and intra-month seasonality. Forecasts beyond the immediate future (e.g., September 1–7) may lack reliability due to the absence of historical context from other months or years.

B. Data Quality and Coverage

A substantial number of records lacked complete metadata, particularly state-level information, due to offshore locations or unidentifiable coordinates. While these entries were excluded or labeled appropriately, their presence indicates incomplete spatial coverage. Moreover, minor errors in ZIP code or geolocation alignment could affect regional aggregation and interpretation.

C. Modeling Constraints

Although SARIMA captured weekly seasonality and short-term structure, it assumes linearity, stationarity (after differencing), and normally distributed residuals [5]. These assumptions were only partially satisfied:

- Residuals exhibited non-normality and early outliers
- The model produced wide confidence intervals, reflecting volatility and potential overfitting to spikes in the training period

Additionally, the model does not incorporate exogenous variables (e.g., temperature, humidity, atmospheric pressure) which are likely drivers of lightning activity.

D. Spatial Modeling Limitations

The analysis did not apply formal spatial autocorrelation models (e.g., Moran's I or spatial regression) or advanced spatial interpolation methods. While interactive maps and choropleths highlighted regional differences, spatial relationships between neighboring areas were not statistically tested or modeled.

E. Generalizability

Findings are specific to August 2018 and may not generalize to other months, years, or geographic contexts. Extreme weather events, such as the high-strike days observed mid-month, may skew statistical patterns and limit the stability of conclusions drawn from a single month's data.

F. Ethical and Societal Considerations

This study is based on publicly available environmental data and does not involve any human subjects or private information, thereby posing minimal ethical risk. However, certain societal limitations should be considered:

- Interpretation risk: Forecasts produced from a short and volatile dataset may lead to overconfident or misleading conclusions if taken out of context. Misuse of such forecasts in policy or media could distort public understanding of weather risk.
- Communication of uncertainty: The wide prediction intervals observed in the SARIMA model highlight the importance of clearly communicating model uncertainty to prevent misinterpretation of deterministic predictions in high-impact domains like disaster preparedness.
- Bias in data availability: The dataset includes only recorded strikes, which are typically better captured in regions with dense monitoring infrastructure. This may inadvertently underrepresent lightning activity in underserved or rural areas, raising concerns about data bias in environmental modeling.

Future studies should prioritize transparent reporting of model assumptions, make uncertainty explicit in forecasts, and ensure equitable access to modeling tools and hazard insights for all communities.

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